

Strontium

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Sr This element is named after Strontian, a Scottish village near where its ore strontianite was discovered. Strontium has a toxic, radioactive isotope called strontium-90. Nuclear accidents, such as the 1986 disaster in Chernobyl, Ukraine, have released a large amount of strontium-90 into the atmosphere.

Signal flare

Compounds of the group 2 elements are used to produce colored flames. Calcium compounds give an orange-red color, while barium compounds produce green. Strontium compounds provide the bright crimson found in fireworks and signal flares (as pictured).



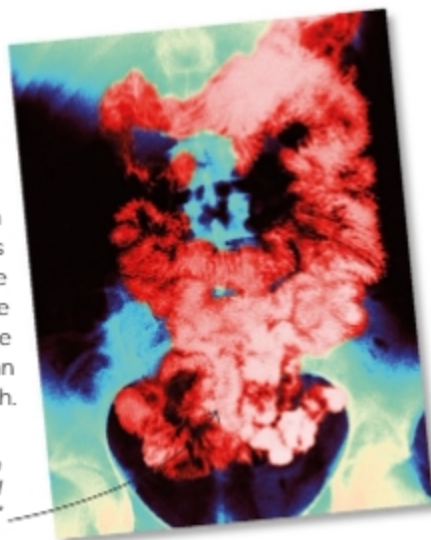
Barium

56
Ba Discovered in 1808 by the English scientist Humphry Davy, the name of this highly reactive metal comes from the Greek word *barys*, meaning "heavy." Interestingly, it is not barium itself that is heavy but the minerals (such as baryte) in which it is found.

Barium meal

In this medical test, a patient swallows a "meal" of barium sulfate, a harmless compound that passes through the digestive tract. This allows a patient's digestive organs—highlighted by the barium compound—to be examined clearly using an X-ray photograph.

X-ray of a healthy human digestive tract, highlighted using a "barium meal."



Radium

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Ra A very radioactive element, radium gets its name from the Latin word *radius*, meaning "ray" (radium glows in the dark). In the early 20th century, radium compounds were used in medications because people believed that radium cured diseases. However, its use was phased out when people who took these drugs became unwell.

This ore contains 0.02 oz (0.7 g) of radium in every 2,205 lb (1,000 kg) of rock.

Uraninite

Formerly known as pitchblende, uraninite is mostly uranium oxide as well as other elements that form as the uranium decays. Polish-born French scientist Marie Curie discovered both radium and polonium by processing huge amounts of uraninite. Polonium was named after Poland.



Glow in the dark

An isotope of radium, radium-226 was used in luminous paints to make watch and clock dials glow in the dark. The paints became popular in the 1920s, long before their health risk was fully understood. Many people, especially women, who worked with this paint became sick and even died as a result of exposure to radium.



The radium in this paint makes the numbers glow green-blue in the dark.

Marie Curie

In 1898, Marie Curie discovered radium and polonium, sharing credit for the former with her husband, Pierre Curie. She received the Nobel Prize in Chemistry in 1911 for these discoveries, becoming the first woman to win a Nobel Prize, and she remains the only woman to have received two.



Medal commemorating the 100th anniversary of Marie Curie's birth, 1967